



1. Draw an example of vertical angles.
2. Draw an example of corresponding angles.

Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Target

- I can use postulates and theorems to find unknown angle measures when parallel lines are intersected by a transversal.



3.3.1 Warm-Up: Using Math in Making Guitars

Patrick Evans is a production manager at Maton Guitars in Australia. Country music star Keith Urban uses Maton guitars. The precision necessary in making each guitar requires mathematics, or “maths,” as Australians call it.

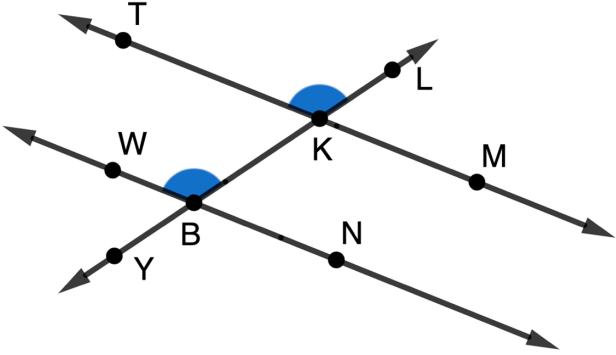
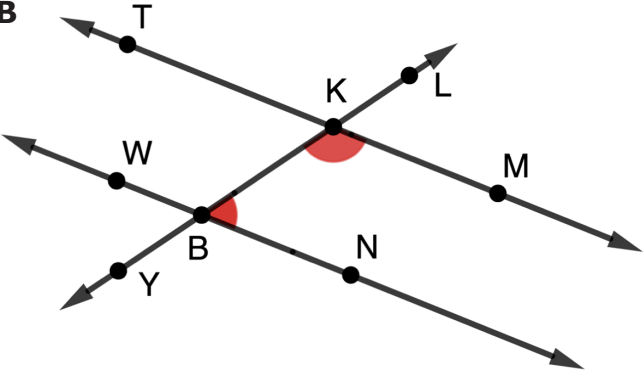


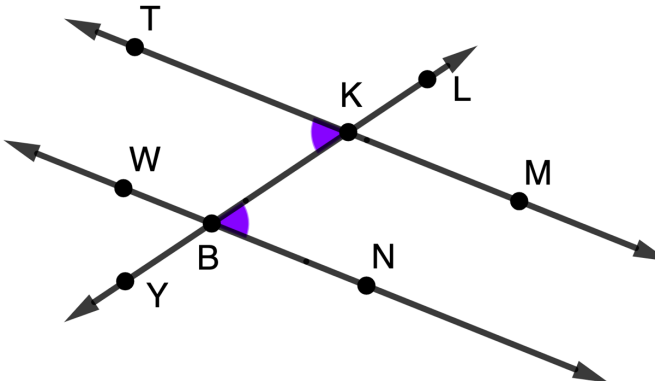
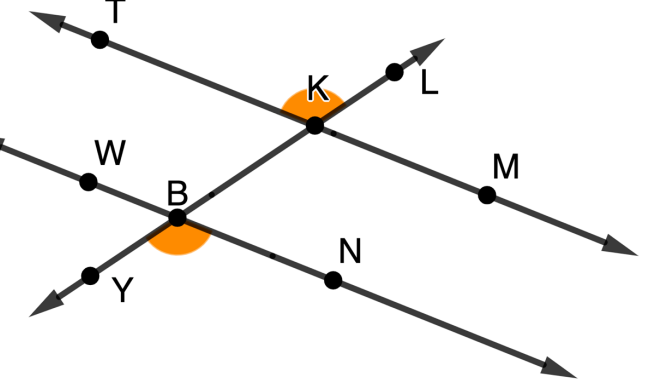
1. What was one thing you found interesting about the process of making a guitar?



3.3.2 Guided Instruction: Angle Relationships

1. In each diagram, $\overline{TM} \parallel \overline{NW}$ where $\overline{YL}$ is a transversal.
 - a. Use patty paper to determine if the angles marked using color are congruent.

Diagram	Congruent?
A 	☐ Yes ☐ No
B 	☐ Yes ☐ No

C 	☐ Yes ☐ No
D 	☐ Yes ☐ No



Alternate Exterior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.

- b. Which diagram shows a pair of angles that are alternate exterior angles?
- ☐ Diagram A
 - ☐ Diagram B
 - ☐ Diagram C
 - ☐ Diagram D

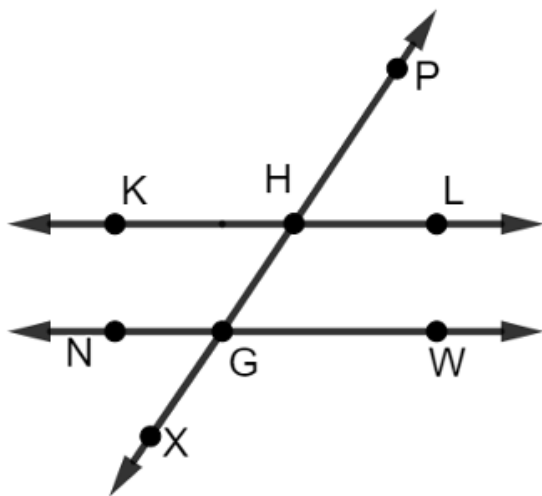


Alternate Interior Angles Theorem

If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.

- c. Which diagram shows a pair of angles that are alternate interior angles?
- ☐ Diagram A
 - ☐ Diagram B
 - ☐ Diagram C
 - ☐ Diagram D

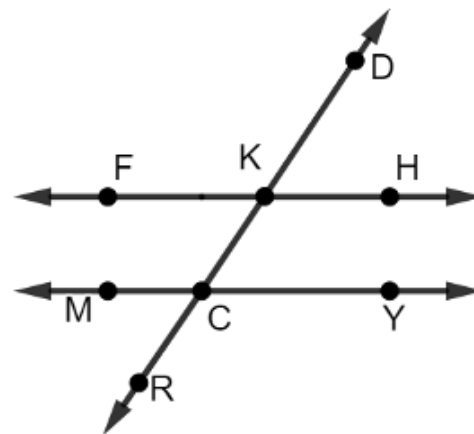
2. $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$, where $\overleftrightarrow{XP}$ is a transversal.



Work with your partner to name each pair of angles as corresponding, alternate exterior, alternate interior angles, or none of these.

Angles	Type
$\angle WGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle LHG$ and $\angle WGX$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle NGX$ and $\angle KHP$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above
$\angle KHG$ and $\angle WGH$	☐ Corresponding angles ☐ Alternate exterior angles ☐ Alternate interior angles ☐ None of the above

3. Given: $\overleftrightarrow{FH} \parallel \overleftrightarrow{MY}$
Prove: $\angle DKH \cong \angle RCM$



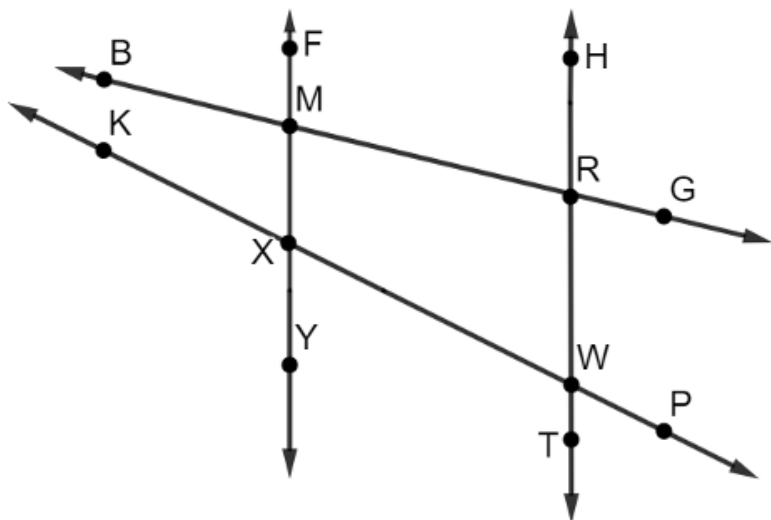
Fill in the table to complete the proof.

Statement	Reason
1.	1. Given
2.	2. Vertical angles are congruent.
3.	3.
4.	4. Transitive Property of Congruence



3.3.3 Try It: Angles and Parallel Lines

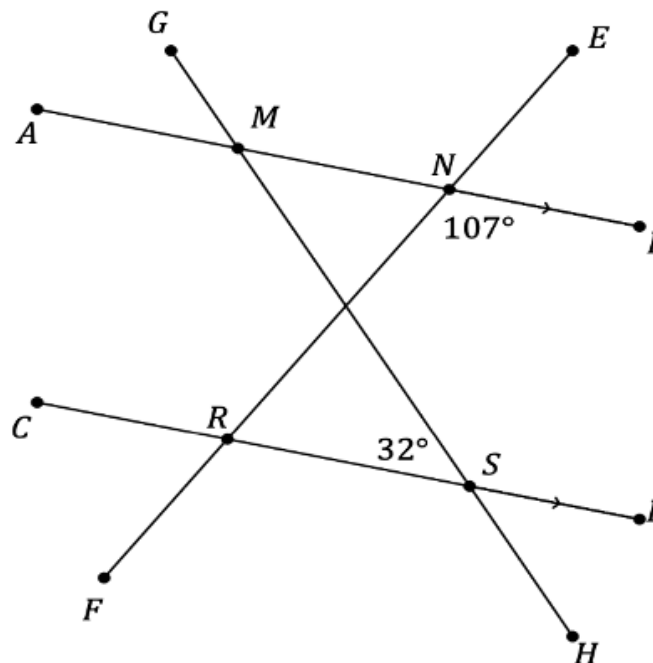
1. $\overleftrightarrow{FY} \parallel \overleftrightarrow{HT}$ where $\overleftrightarrow{BG}$ and $\overleftrightarrow{KP}$ are transversals and $m\angle BMF = 76^\circ$ and $m\angle KXY = 117^\circ$.



Complete the table.

Angle	Angle Measure
$\angle MRW$	
$\angle PWT$	
$\angle YXW$	
$\angle RMX$	
$\angle PWR$	

2. $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ where $\overleftrightarrow{GH}$ and $\overleftrightarrow{EF}$ are transversals and $m\angle BNR = 107^\circ$ and $m\angle RSM = 32^\circ$.



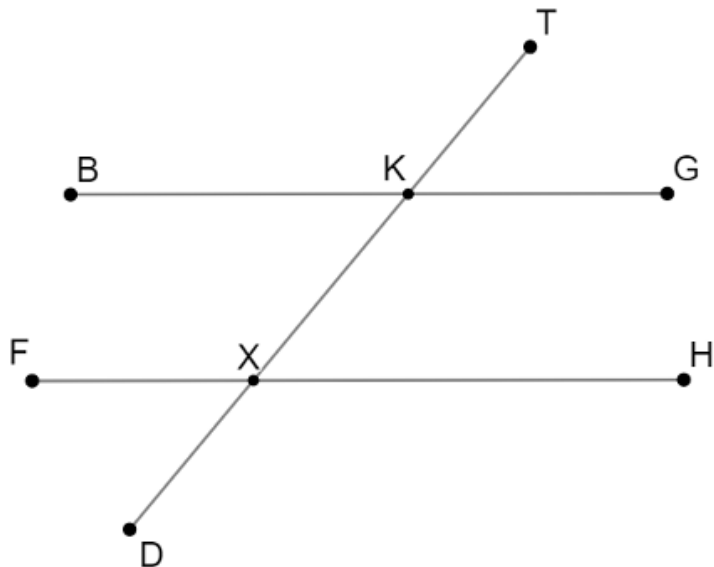
Complete the table.

Angle Measure	Postulate or Theorem Used
$m\angle NRC =$	
$m\angle BMH =$	
$m\angle DRF =$	
$m\angle GMA =$	



3.3.4 Your Turn!

1. $\overline{BG} \parallel \overline{FH}$ where $\overline{TD}$ is a transversal and $m\angle HXD = 134^\circ$.



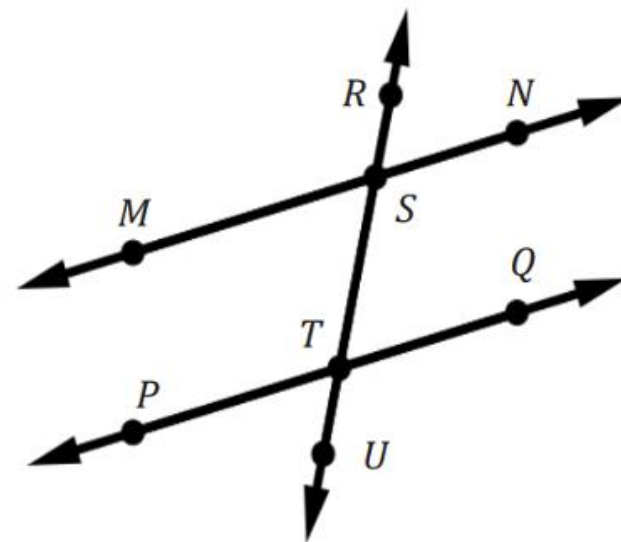
Find the measure of each angle. Then, state the angle relationships used to determine the value.

Angle Measure	Angle Relationships
$m\angle GKK =$	
$m\angle FXK =$	
$m\angle BKT =$	



3.3.5 Wrap-Up: Ticket Out the Door

1. $\overleftrightarrow{MN} \parallel \overleftrightarrow{PQ}$ and $m\angle PTS = 100^\circ$.



- In relation to the parallel lines, what angle relationship do $\angle TSN$ and $\angle PTS$ have?
- What is $m\angle TSN$?